Yubo Wang (王钰博)

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Research Interests

Agentic Memory, Multimodal, Self-evolving Agents, Agentic RL.

Education

2021 – Present **Ph.D. Student/Candidate, CSE Department, HKUST**
Supervised by Prof. Lei Chen.
Work closely with Dr. Haoyang Li, Dr. Shimin Di and Dr. Hao Xin

2019 – 2021 **M.Sc. Computer Science, CISE Department, University of Florida**.
Supervised by Prof. Eric Jing Du

2015 – 2019 **B.Sc. Computer Science and Technology, CCST Department, Jilin University**.
Supervised by Prof. Fengfeng Zhou

Internship Experiences

- **Pailitao, Taobao & Tmall Group, Alibaba Inc.**
  Beijing, 2026.4-Present
  Research on multimodal agentic search and long-term memory systems. Responsible for RL post-training of memory-manager LLMs (The CMI-MEM-4B LLM checkpoint is available at ModelScope) and multimodal agentic memory searcher.

1st Author Publications

- **CMI-Mem: Toward Generalizable Long-Term Memory Management via CMI-Augmented Reinforcement Learning (In Submission)**
  Work during internship at Taobao & Tmall Group. Developed an Agentic RL framework that trains memory-manager LLMs with conditional-mutual-information rewards, improving OOD generalizability.
- **Demory: Training LLMs for On-Demand Memory Invocation via Agentic RL (Ongoing)**
  Work during internship at Taobao & Tmall Group. Developing an Agentic RL framework that trains a 9B searcher LLM over data created from CMI-Mem generated memory banks, with fine-grained rewards for on demand memory retrieval and query rewriting.
- **MGRAG: Semantic Subgraph Matching and Graph-Aware Caching for Multimodal Retrieval-Augmented Generation (VLDB 26 (CCF-A))**
  Built a multimodal graph RAG system for fine-grained visual understanding on text-image corpora, with 2-staged adaptive multimodal retrieval and visual structural aware multimodal KV cache acceleration.
- **GORAG: Graph-based Online Retrieval Augmented Generation for Dynamic Few-shot Social Media Text Classification (WWW 26 Oral (CCF-A))**
  Model text-label correlations as graph structured memory. Using weighted edges to represent the reliablity of certain memory, and dynamically update the graph memory with new information to incorporate streaming trustworthy data.
- **AGRAG: Advanced Graph-based Retrieval-Augmented Generation for LLMs (ICDE 26 (CCF-A))**
  Design a semantic guided graph expansion algorithm based on Steiner Tree, to adaptive determin the optimal subgraph for retrieval, and achieve better performance on summarization queries.
- **KGLink: A Column Type Annotation Method that Combines Knowledge Graph and Pre-trained Language Model (ICDE 24 (CCF-A))**
  Leverages tabular structures to filter and refine retrieved Knowledge Graph (KG) information. Carry out end-to-end training for tabular understanding tasks.
- **Understanding the Embedding Methods on Hyper-relational Knowledge Graph (CIKM 25 (CCF-B))**